

Walking the Hunter Biped on Agricultural Soil

Two experiments: rigid furrows (deploys) vs. deformable soil (a clean null)

HumanoidVerse · Hunter · IsaacSim/IsaacLab

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The question, in one slide

Goal: make the Hunter biped walk across a tilled crop field in simulation, robustly enough to deploy.

We ran **two separate experiments**, each asking a different question:

1. Furrows (geometry)

Can the robot handle the *shape* of plowed ground — ridges and grooves — if the ground is still *rigid*?

Answer: **works** — produced the policies we deploy.

2. DFH (deformable soil)

Does adding *soft, sinking, dragging* soil physics make a *measurably better* walker than rigid training?

Answer: **null** — soil-trained did not beat rigid-trained.

Both share the same robot (Hunter), simulator (IsaacSim 4.2), and algorithm (PPO-ROA, 234-dim history observation).

Experiment 1: Furrows — what is being tested

A **furrow** = the groove + crest pattern left by plowing. We bake that shape into a *rigid* heightfield and ask the robot to walk across the rows.

- Ground is hard (friction $\mu_s=0.60$, $\mu_d=0.45$); only the **geometry** is hard, not the material.
- Trained as a **3-stage curriculum** — start easy, promote when stable, then harden the terrain.

Stage	Groove depth	Row spacing	Row tilt	Crest lift
1 Easy	0.08–0.12 m	2.7–3.2 m	0°	0.05 m
2 Medium	0.15–0.20 m	2.0–2.5 m	$\pm 3^\circ$	0.03 m
3 Full	0.20–0.30 m	1.5–2.0 m	$\pm 5^\circ$	0.00 m

Each stage = deeper grooves, tighter rows, more misalignment — i.e. progressively harder footing.

Experiment 1: promotion rule & result

We only advance a stage when the policy is genuinely stable:

Promotion	Episode length	Termination reward
Stage 1 $\rightarrow$ 2	≥ 16 s	$\geq -0.05/s$
Stage 2 $\rightarrow$ 3	≥ 18 s	$\geq -0.03/s$ (+ real gait)
Stage 3 target	$\sim 19\text{--}20$ s	stable tracking

Outcome **works**

The furrows curriculum produced the **working walkers we ship**:

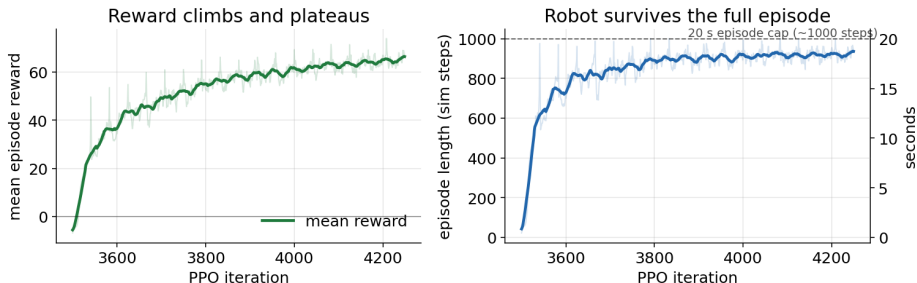
- `v7_baseline` — flat-ground walker (warm-start root)
- `rigid_walker` — walks rigid furrows
- `mildsoil_walker` (model_5050) — **deployment policy**, velocity error 0.098, drives the agri-field demo

All published to HuggingFace ([anhri/n/hunter-dfh-locomotion](https://huggingface.co/anhri/n/hunter-dfh-locomotion)).

Experiment 1: does it actually learn? (yes)

Two signals, read straight off the training logs of the deployment model:

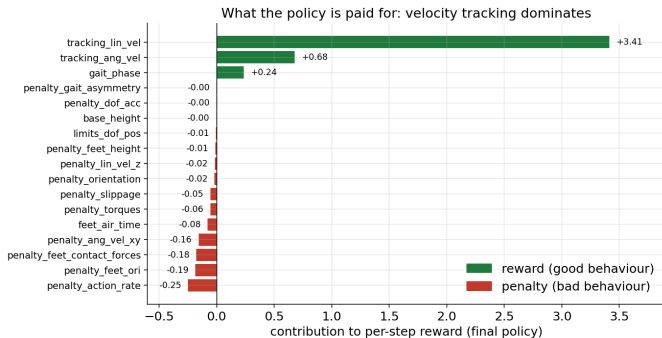
Experiment 1 — Rigid furrows: the walker learns (deployment model v7)



- **Left:** mean reward rises from below zero and *plateaus high* — a healthy PPO curve.
- **Right:** episode length climbs to near the **20s cap** (~1000 steps) — the robot survives the whole episode.
- **Reward up + episodes long = it learned to walk**, not just to stand.

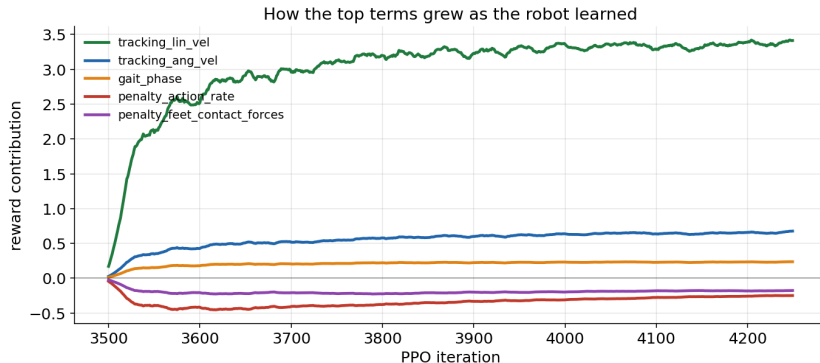
Which rewards actually drove the learning?

The total reward is a sum of **17 terms**. Breaking the final policy apart shows what the robot is really being paid for:



- **Velocity tracking** (tracking_lin_vel, +3.41) dwarfs everything — “go at the commanded speed” is the goal that pulls the policy.
- tracking_ang_vel (+0.68) and gait_phase (+0.24) shape *turning* and a *rhythmic gait*.
- All penalties (red) stay **small** — they tidy up the motion (smooth actions, gentle contacts) without overpowering the goal.

Same story, over time: tracking is the engine



- The green `tracking_lin_vel` curve has **the same shape as the total-reward curve** — it *is* what drove the climb.
- Gait and turning rewards grow gently, then settle into a stable rhythm.
- Penalties (red/purple) deepen only slightly — the cost of moving faster, not a fight against the reward.
- **Takeaway:** learning = “chase the velocity command cleanly”; the other 16 terms are guardrails, not the driver.

Experiment 2: DFH — adding real soil physics

DFH = Deformable Furrowed Heightfield. On top of the furrow shape, the ground now *responds* to the feet. Five physical effects, all GPU-resident:

- 1 **Sinkage** (Bekker): feet press in under load.
- 2 **Slip-sinkage** (Wong–Reece): a dragging foot digs deeper.
- 3 **Plastic ratchet**: a dent stays dented within an episode.
- 4 **Bulldozing**: pushed soil rises in neighbour cells.
- 5 **Anisotropic friction**: harder to slide *across* rows than *along* them.

Difficulty rises with the curriculum: softer soil (lower Bekker moduli), wider anisotropy, deeper sink cap (5 cm → 10 cm).

The force on the robot

Each foot feels a tangential drag:

$$F_{\text{drag}} = k \cdot \text{sinkage} \cdot F_{\text{normal}}$$

clamped at `max_drag_force_n`.

Deeper sink $\Rightarrow$ more drag $\Rightarrow$ the robot must actively fight the soil.

Experiment 2: a fair test — the paired protocol

To claim “soil training helps” we must beat the obvious baseline. So for each random seed we trained **three matched policies**:

Policy	Trained on
RIGID	rigid furrows (no soil forces) — the baseline
SHADOW	soil computed but <i>forces off</i> (sanity check)
DFH_FULL	soil + drag forces active

All three are then evaluated *on the same deformable terrain*. The headline number is the **paired regret**:

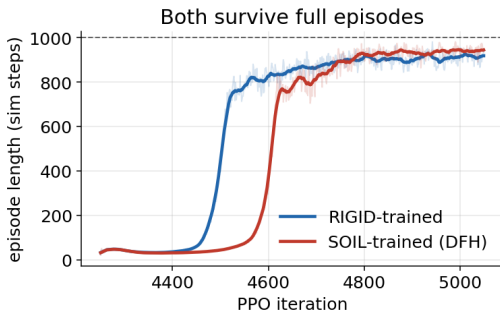
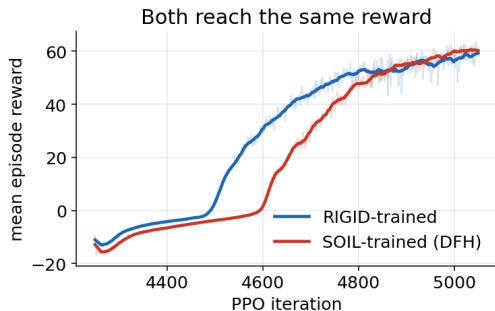
$$\text{regret} = \text{ep_len}(\text{DFH_FULL@soil}) - \text{ep_len}(\text{RIGID@soil})$$

- **Positive** regret $\Rightarrow$ soil training paid off.
- Reward shaping for sink held at **zero** during the test, so any gain is from *dynamics*, not hand-tuned rewards.
- Success bar set *in advance*: drag 5–10% body weight, regret > 0 on every seed, mean regret $\geq 10\%$ of baseline.

Experiment 2: both policies learn (this is not a training failure)

RIGID = trained on the *rigid furrowed heightfield* — the hard, plowed-shape ground from Experiment 1 (geometry only, the surface never moves). **SOIL (DFH)** = the *same* furrows, but the ground now sinks, drags and deforms. So “rigid” is the furrow heightfield with the soil physics switched off.

Experiment 2 — Both policies learn equally well; soil training adds no edge

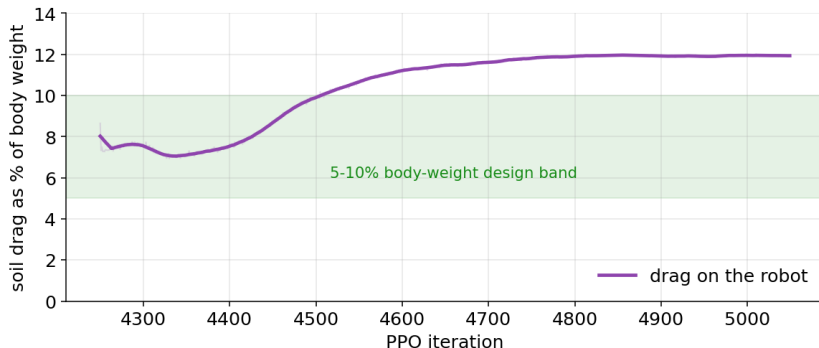


- Both climb to the **same reward** (~ 60) and **same episode length** (~ 1000 steps).
- The soil-trained one takes off a bit later (it fights drag) but **catches up completely** — so any later gap is *not* a training failure.

Experiment 2: the soil load is real — and it copes

Drag as % of body weight = the backward pull the soil puts on the feet, divided by the robot's own weight. We normalise by weight so the number is intuitive and robot-independent: “10%” means the soil resists with a force equal to one-tenth of the robot — like wading through mud. It is our honest measure of *how hard the terrain is fighting the robot*.

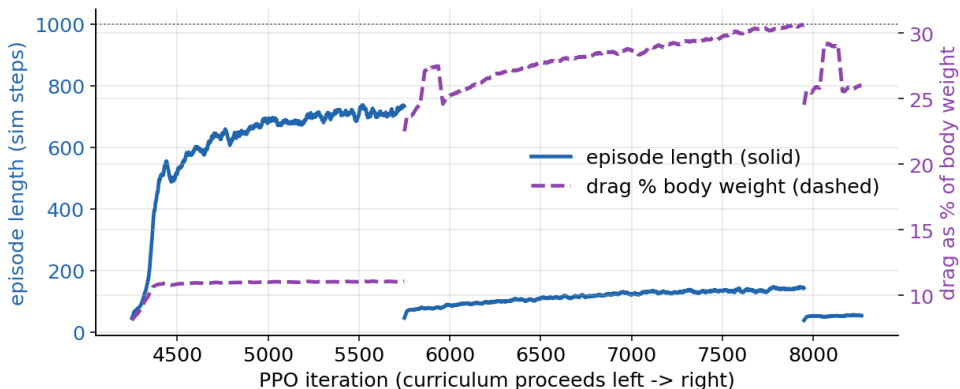
Experiment 2 — A real load (~12% body weight) the robot learns to walk through



- Settles at **~12% body weight** — a genuine, persistent load, in the range we designed for.
- Episodes still reach the cap, so the robot **learned to walk through that drag**: the physics is doing real work on it.

Experiment 2: why the harder soil stages are brutal

Why deeper soil is brutal: as drag rises (8% -> 30% BW), survival collapses



- As the curriculum deepens the soil, drag rises from $\sim 11\%$ to $\sim 25\text{--}30\%$ body weight (dashed).
- Survival (solid) **collapses** from ~ 730 steps to $\sim 50\text{--}150$ — a third of body weight in drag is more than the gait can absorb.
- This is why we report results on the mild/tuned regime, not full/wet soil.

Experiment 2: the result — a clean null

Eval on soil	DFH_FULL	RIGID	Regret (DFH-RIGID)
Tuned ($k=40$)	26.2 s	27.0 s	-0.8 s
De-saturated ($k=12$)	27.6 s	31.4 s	-3.8 s

- Rigid-trained policies **matched or beat** soil-trained ones on the soil itself — the opposite of the hypothesis.
- First suspect: the drag cap was saturating (clip 17–31%), maybe hiding a real signal. So we **removed the confound** — lowered k until clipping fell to 3.2%.
- The null got *stronger*, not weaker ($-0.8 \rightarrow -3.8$ s). The gap is real, not an artifact.

Verdict **null**

Soil-aware training did **not** produce a better soil walker. We report this as an honest negative result rather than overclaim.

Why we trust the null (and what *did* work)

The null is credible:

- Negative control passed: DFH_FULL beat SHUFFLED (+2.3 s) — the forces *are* doing something coherent.
- Confound eliminated: clipping cut to 3.2%, mean drag only $\sim 2.4\%$ body weight — yet rigid still won.
- Trend is consistent across both dose levels.

Independent multi-round review scored the DFH *claim* 3.0/10 (evidence 2.0) and recommended pivoting to the honest null — which is what these slides do.

The engineering succeeded:

- DFH layer attaches & runs live in IsaacSim, 120×120 grid, writeback every 4 steps.
- No measurable speed cost (~ 0.78 s/iter).
- 27/27 unit + integration tests pass.

So the negative is a *scientific* finding, not a broken pipeline.

Takeaways

- ① **Furrow geometry is enough to deploy.** A staged curriculum on rigid plowed terrain gives a stable Hunter walker — `mildsoil_walker` powers the agri-field demo.
- ② **Deformable-soil training did not help.** Under a pre-registered, confound-controlled test, soil-aware policies did not beat rigid-trained ones on soil. Clean null.
- ③ **The DFH simulator itself is solid** — correct, fast, tested — and remains a reusable IsaacLab extension for future, better-calibrated studies.

Models: `huggingface.co/anhrisn/hunter-dfh-locomotion` · Code: `extensions/dfh/`

Thank you

Questions on the furrows curriculum or the DFH null welcome.